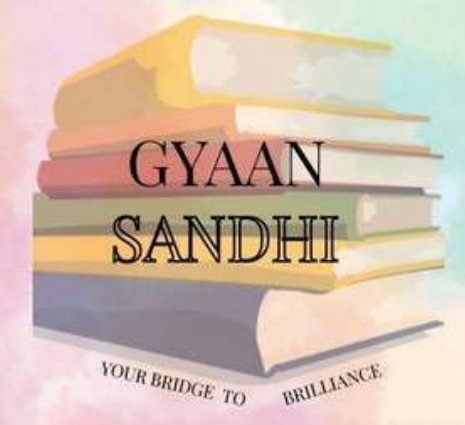
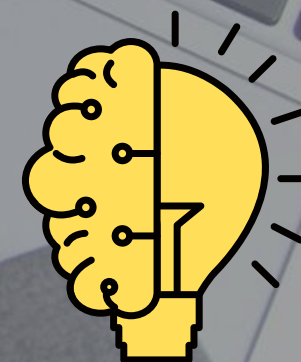




UNIT 4

Data Preparation

Business Intelligence



Data Preprocessing

- Data preprocessing is the process of cleaning, transforming, and preparing raw data before analysis.
- It ensures data is accurate, consistent, and well-structured.
- Preprocessed data improves the quality of analysis and decision-making.

Need for Data Preprocessing

- Data is collected from multiple sources such as databases, sensors, and files.
- Raw data may contain:
 - Missing values
 - Duplicate records
 - Inconsistent formats
 - Errors and noise
- Poor-quality data can lead to incorrect results and unreliable reports.
- Data preprocessing helps improve data quality and reliability

Data Preprocessing

Benefits of Data Preprocessing

- Cleans and organizes data.
- Handles missing and duplicate values.
- Converts data into a consistent format.
- Improves accuracy of analysis and machine learning models.
- Reduces errors and enhances data usability.

Data Preprocessing Techniques – Data Cleaning

- Removes incorrect, incomplete, duplicate, or irrelevant data.
- Identifies and handles missing values.
- Detects and removes outliers.
- Ensures consistency in column names and labels.
- Example: Replacing missing prices with the average price.

Data Preprocessing

Data Preprocessing Techniques – Data Transformation

- Converts data into a suitable format for analysis.
- Standardizes data structures and formats.
- Makes data from multiple sources compatible.
- Improves consistency and integration of datasets.
- Example: Converting all date formats to DD-MM-YYYY.

Data Cleaning

- Data cleaning is the process of removing or correcting inaccurate, incomplete, duplicate, or irrelevant data from a dataset.
- It improves the quality and reliability of data.
- Clean data leads to more accurate analysis and reporting.

Need for Data Cleaning

- Real-world data is collected from various sources such as databases, files, and sensors.
- Raw data may contain:
 - Errors
 - Missing values
 - Duplicate records
 - Incorrect formats
- Unclean data can produce inaccurate results and poor business decisions.
- Data cleaning ensures data accuracy, consistency, and reliability.

Data Cleaning

Benefits of Data Cleaning

- Improves data quality.
- Increases accuracy of analysis.
- Reduces errors and inconsistencies.
- Enhances decision-making.
- Produces reliable reports and insights.

Method 1 – Removing Duplicates

- Identifies and removes repeated records from a dataset.
- Prevents redundant data and incorrect analysis.
- Ensures each record appears only once.
- Example: If the same customer is listed twice, keep only one record.

Data Cleaning

Method 2 – Handling Missing Values

- Detects missing or incomplete data.
- Missing values can be:
 - Replaced using Mean, Median, or Mode.
 - Removed if too many values are missing.
- Helps maintain data completeness and accuracy.
- Example: Fill missing product prices with the average price.

Method 3 – Standardizing Formats

- Converts data into a uniform and consistent format.
- Ensures compatibility across datasets.
- Common examples:

Data Cleaning

- Date formats (DD-MM-YYYY)
- Currency formats (INR)
- Example: Convert "Jan 1, 2025" to "01-01-2025".

Method 4 – Outlier Detection and Handling

- Identifies unusual or extreme values in the dataset.
- Outliers can affect analysis and predictions.
- Such values can be corrected, removed, or investigated further.
- Example: A product price entered as ₹1,00,000 instead of ₹1,000 can be flagged and corrected.

Data Validation

- Data validation is the process of checking whether input data is correct, meaningful, and useful before it is used for analysis or reporting.
- It ensures data follows predefined rules and formats.
- Validation helps prevent errors and improves data accuracy.
- Examples:
 - Correct date formats
 - Valid values within a range
 - No missing mandatory fields

Importance of Data Validation

- Prevents incorrect data from entering the system.
- Improves data quality and reliability.
- Reduces errors in reports and analysis.
- Ensures consistency and accuracy of business decisions.

Incompleteness

- Incomplete data occurs when some values are missing from a dataset.
- Causes of incomplete data:
 - Human errors
 - Data collection issues
 - System failures
- Missing information can affect the accuracy of analysis and reporting.
- It should be handled by:
 - Filling missing values
 - Removing incomplete records when necessary

Noise in Data

- Noise refers to random, irrelevant, or meaningless data that does not follow the actual pattern of the dataset.
- It reduces the quality of analysis and reports.
- Noise can hide important trends and patterns.

Example:

- Sensor errors generating incorrect readings.
- Typographical mistakes in data entry.
- Noise should be identified and removed during data preprocessing.

Inconsistency in Data Quality

- Inconsistency occurs when data is not uniform or standardized.
- Different formats for the same information can create confusion.
- Examples:
 - Dates stored as "01-01-2025" and "Jan 1, 2025"
 - Product names written as "TV" and "Television"
- Inconsistent data can lead to incorrect analysis and conclusions.

Effects of Poor Data Quality

- Inaccurate reports and insights.
- Wrong business decisions.
- Reduced reliability of analysis.
- Increased data processing errors.
- Difficulty in integrating data from multiple sources.

Data Transformation

- Data transformation is the process of converting raw data into a clean and usable format.
- It prepares data for reporting, dashboards, and analysis.
- The goal is to make data consistent, understandable, and suitable for processing.

Need for Data Transformation

- Raw data is often collected from different sources such as databases, CSV files, and APIs.
- Data may exist in different formats and structures.
- Inconsistent data makes analysis difficult.
- Data transformation helps by:
 - Reorganizing data
 - Converting formats
 - Standardizing values

Data Transformation

Example of Data Transformation

- Consider customer birthdates stored in different formats:
- 01/01/2025
- Jan 1, 2025
- 2025-01-01
- These formats are inconsistent.
- Data transformation converts them into a standard format such as:
- 01-01-2025 (DD-MM-YYYY)
- This ensures consistency and easier analysis.

Process of Data Transformation

- Data StorageStore or load the transformed data into a data warehouse for analysis.

Data Transformation

- Data Inspection :- Identify inconsistencies, errors, and formatting issues.
- Apply Transformation Rules :- Convert formats, change data types, and standardize values.
- Data Storage :- Store or load the transformed data into a data warehouse for analysis.

Technique 1 – Normalization

- Normalization scales numerical data to a common range, usually between 0 and 1.
- It removes the effect of different units and value ranges.
- Improves the performance of data analysis and machine learning models.

Example:

- Income values: ₹10,000, ₹1,00,000, ₹5,00,000
- After normalization: 0.01, 0.20, 1.00

Data Transformation

Technique 2 – Encoding (Categorical to Numerical)

- Encoding converts categorical or text data into numerical values.
- Numerical representation is required for calculations and machine learning algorithms.
- Makes data easier to process and analyze.

Example:

- Yes $\rightarrow$ 1
- No $\rightarrow$ 0

Technique 3 – Data Aggregation

- Data aggregation groups data and calculates summary statistics.

Data Reduction

- Data Reduction is the process of reducing the amount of data while maintaining its meaning and usefulness.
- It helps manage large datasets efficiently.
- Reduces storage, processing time, and computational costs.
- Makes data analysis faster and more focused.
- Example:
 - A dataset contains 100 columns.
 - Only 10 columns are relevant for sales analysis.
 - Keeping only those 10 columns simplifies analysis.

Need for Data Reduction

- Large datasets require more storage space.

Data Reduction

- Processing large volumes of data can be slow.
- Complex data increases computational costs.
- Data reduction improves efficiency without losing important information.
- Helps in faster decision-making and analysis.

Technique 1 – Sampling

- Sampling involves selecting a representative subset from a large dataset.
- The sample reflects the characteristics of the original dataset.
- Reduces the amount of data to be processed.
- Saves time and computing resources.
- Example:
 - Analyze 10,000 customer records from a dataset of 1 million customers.

Data Reduction

Technique 2 – Feature Selection

- Feature selection identifies the most relevant attributes (columns) in a dataset.
- Irrelevant or redundant features are removed.
- Reduces data dimensionality.
- Improves model performance and interpretability.
- Example:
- For sales prediction, keep features such as Age, Income, and Purchase History while removing unnecessary columns.

Technique 3 – Principal Component Analysis (PCA)

- PCA is a mathematical technique used to reduce high-dimensional data.
- Converts original features into a smaller set of new variables called Principal Components.

Data Reduction

- Principal components capture most of the important information in the data.
- Useful for datasets with many correlated variables.

Applications:

- Finance
- Healthcare
- Image Processing
- Data Analytics

Benefits of PCA

- Reduces dataset complexity.
- Removes redundancy among variables.
- Improves analytical and machine learning performance.
- Enables easier visualization in 2D or 3D.

Data Reduction

Dimensionality Reduction

- Dimensionality Reduction decreases the number of input variables while preserving important information.
- Helps simplify data structures and analysis.

Problems with High-Dimensional Data:

- Overfitting
- Increased storage costs
- Slower processing
- Difficult visualization

Benefits:

- Faster processing
- Better visualization

Data Reduction

- Improved model performance
- Reduced complexity

Types of Dimensionality Reduction

Supervised Dimensionality Reduction

- Uses output labels while reducing dimensions.
- Focuses on features that influence the target variable.

Unsupervised Dimensionality Reduction

- Does not use output labels.
- Finds hidden patterns and structures in data.

Data Reduction

Data Compression

- Data Compression reduces the physical size of a dataset.
- Saves storage space and speeds up data transmission.
- Focuses on efficient storage and data handling.

Types of Data Compression

Lossless Compression

- Original data can be fully recovered.
- No information is lost.
- Examples: ZIP files, PNG images

Lossy Compression

- Some information is permanently removed.
- Achieves higher compression ratios.
- Examples: JPEG images, MP3 audio files

Data Discretization

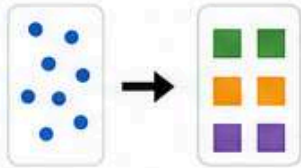
1 Data Discretization

- Data Discretization is the process of converting continuous data into discrete groups or intervals.
- It simplifies data and makes analysis easier.
- Commonly used in Business Intelligence (BI) and Data Mining.
- Helps identify patterns and improves algorithm performance.



Example:

- Exact Ages: 23, 24, 25, 26...
- Discretized Groups:
 - 18–25 = Young
 - 26–35 = Adult
 - 36–60 = Senior



2 Need for Data Discretization

- Simplifies complex continuous data.
- Makes patterns easier to understand.
- Reduces data complexity.
- Improves efficiency of data mining and machine learning algorithms.
- Enhances visualization and reporting.

3 Methods of Data Discretization

1. Binning



- One of the most common discretization techniques.
- Divides continuous values into a fixed number of intervals (bins).
- Each bin contains a range of values.
- Helps reduce noise and simplify analysis.

4 Types of Binning – Equal Width Binning

- The entire data range is divided into intervals of equal size.
- Each bin has the same width.

Example:

- Data Range: 0–100
- Number of Bins: 5



- Simple to implement.
- Easy to understand.

5 Types of Binning – Equal Frequency Binning

- Each bin contains the same number of data points.
- Bin widths may vary depending on data distribution.

Example:

- Dataset contains 10 values.
- Divide into 2 bins.
- Each bin contains 5 values.

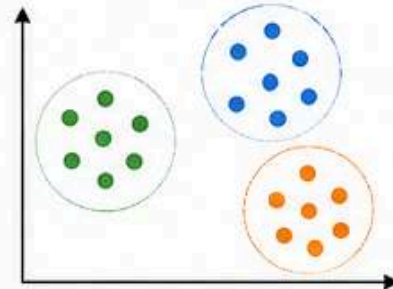


Advantages:

- Balances the number of observations in each group.
- Useful for skewed datasets.

6 Types of Binning – Clustering-Based Binning

- Groups similar values using clustering algorithms.
- Commonly uses the K-Means algorithm.
- Values within the same cluster are treated as one group.

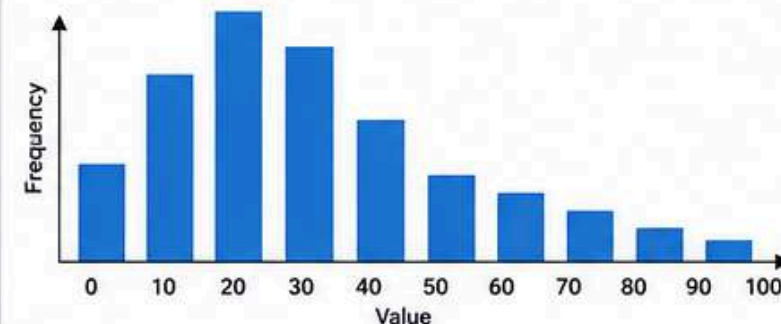


Advantages:

- Creates meaningful groups based on data characteristics.
- Adapts to the natural structure of data.

7 Histogram-Based Discretization

- Uses a histogram to determine interval boundaries.
- Divides data based on its frequency distribution.
- Creates bins according to natural gaps in the data.



Example:

- Most values lie between 0–30.
- More bins are created in this range.
- Fewer bins are created where data is sparse.

8 Advantages of Histogram-Based Discretization



Reflects the actual distribution of data.



Identifies natural groupings.



Provides better representation than simple fixed-width bins.



Useful for exploratory data analysis.

9 Benefits of Data Discretization



Simplifies continuous data.



Reduces data complexity.



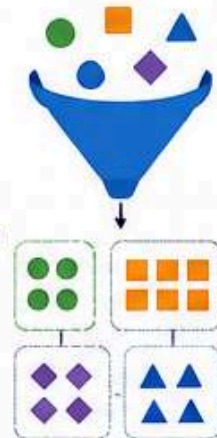
Improves pattern recognition.



Enhances performance of data mining and machine learning algorithms.



Makes reports and visualizations easier to interpret.



10 Conclusion



Data Discretization converts continuous data into meaningful categories or intervals.



Common methods include:

- Binning
 - Equal Width Binning
 - Equal Frequency Binning
 - Clustering-Based Binning



• Histogram-Based Discretization



It helps simplify data, improve analysis, and enhance decision-making.



Data Exploration

- Data Exploration is the process of understanding the structure, patterns, and key characteristics of a dataset before performing detailed analysis.
- It helps analysts become familiar with the data.
- Also known as Exploratory Data Analysis (EDA).
- It is an important step in Business Intelligence (BI) and Data Analytics.

Purpose of Data Exploration

- Understand the nature of the data.
- Identify data quality issues.
- Discover patterns and trends.
- Detect outliers and anomalies.
- Support better business decisions.

Data Exploration

What is Analyzed During Data Exploration?

- 1. Types of Data
- Numerical data
- Text data
- Categorical data

2. Data Distribution

- Mean
- Median
- Mode
- Range

3. Data Quality

- Missing values
- Duplicate records

4. Relationships and Outliers

- Correlation between variables
- Detection of unusual values

Data Exploration

Techniques Used in Data Exploration

Summary Statistics

- Mean
- Median
- Mode
- Minimum and Maximum Values
- Standard Deviation

Data Visualization

- Histograms
- Bar Charts
- Scatter Plots
- Pie Charts
- Line Charts

Example – Superstore Sales Dataset

Suppose we are analyzing a Superstore Sales dataset in Microsoft Power BI.

Important Columns:

- Order ID
- Customer Name
- Region
- Sales
- Profit
- Shipping Date

The first step is to understand the meaning and content of each column

Data Exploration

Checking Summary Statistics

- Calculate average sales.
- Find maximum and minimum profit.
- Analyze total revenue and profit.

Example:

- Average Sales = ₹5,000
- Maximum Profit = ₹15,000

These statistics provide a quick overview of business performance.

Detecting Missing Values

- Identify columns with missing information.
- Missing values can affect analysis results.

Example:

- Some records have missing Shipping Date values.

Possible actions:

- Fill missing values.
- Remove incomplete records.

Data Exploration

Visualizing Data Patterns

Bar Chart

- Shows Technology category has the highest sales.

Scatter Plot

- Reveals that higher sales often lead to higher profits.
- Helps identify trends and relationships.

Visualizations make complex data easier to understand.

Spotting Outliers

- Outliers are unusually high or low values.
- They may indicate errors or special cases.

Example:

- Some orders have very high sales but zero profit.
- Such records should be investigated further.

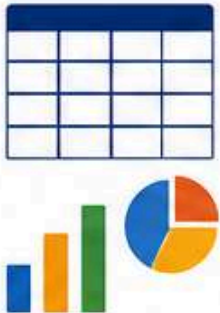
Benefits:

- Improves data quality.
- Prevents misleading analysis.

Contingency Table & Marginal Distribution

1 Contingency Table

- A Contingency Table is a statistical table used to display the relationship between two or more categorical variables.
- It shows the frequency (count) of observations for each combination of categories.
- Helps identify patterns, trends, and relationships in data.
- Commonly used in surveys, business reports, and data analysis.



2 Purpose of a Contingency Table

- Organizes categorical data in a structured format.
- Helps compare different categories.
- Reveals possible relationships between variables.
- Supports decision-making through data analysis.

3 Example of a Contingency Table

Relationship between Gender and Product Preference

Gender	Electronics	Clothing	Total
Male	20	10	30
Female	15	25	40
Total	35	35	70

- Observations:
- More males prefer Electronics.
 - More females prefer Clothing.
 - Total respondents = 70.



4 Benefits of Contingency Tables

- Easy to understand and interpret.
- Summarizes large amounts of categorical data.
- Helps identify associations between variables.
- Useful in market research and business intelligence.

5 Marginal Distribution

- Marginal Distribution refers to the totals of rows or columns in a contingency table.
- It shows the distribution of a single variable while ignoring the effect of other variables.
- These totals appear at the margins (edges) of the table.
- Provides a summary view of the data.

Gender	Product Preference		Total
	Electronics	Clothing	
Male	20	10	30
Female	15	25	40
Total	35	35	70

Marginal Totals

6 Marginal Distribution of Gender

Using the previous contingency table:

Gender	Total
Male	30
Female	40
Total	70



- Interpretation:
- Male respondents = 30
 - Female respondents = 40

This represents the marginal distribution of Gender.

7 Marginal Distribution of Product Preference

Using the previous contingency table:

Product Preference	Total
Electronics	35
Clothing	35
Total	70



- Interpretation:
- Electronics preferred by 35 respondents.
 - Clothing preferred by 35 respondents.

This represents the marginal distribution of Product Preference.

8 Importance of Marginal Distribution

- Provides an overall summary of each variable.
- Helps calculate probabilities and percentages.
- Simplifies data interpretation.
- Useful for comparing category frequencies.

9 Applications

- Contingency Tables
- Survey Analysis
 - Market Research
 - Customer Preference Analysis
 - Business Reporting

- Marginal Distributions
- Probability Calculations
 - Statistical Analysis
 - Data Summarization
 - Decision-Making

10 Conclusion

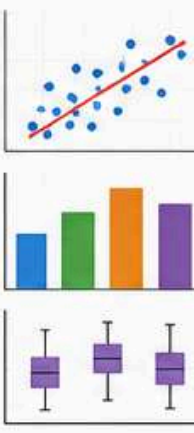
- A Contingency Table displays the relationship between categorical variables using frequency counts.
- A Marginal Distribution summarizes the totals of rows or columns in a contingency table.
- Together, they help analyze relationships, identify trends, and support data-driven decision-making.



Bivariate Analysis and MultiVariate Analysis

1 Bivariate Analysis

- Bivariate Analysis is the analysis of two variables to understand the relationship or association between them.
- It helps identify how one variable changes with respect to another.
- Commonly used for comparison and correlation studies.
- Techniques include:
 - Scatter Plots
 - Correlation Analysis
 - Bar Charts
 - Box Plots



2 Example of Bivariate Analysis

Marketing Spend vs Sales Revenue

- A business may analyze the relationship between marketing expenditure and sales revenue.
- If marketing spending increases and sales also increase, there is a positive relationship between the variables.

Example:
Higher advertising budget
→ Higher sales revenue



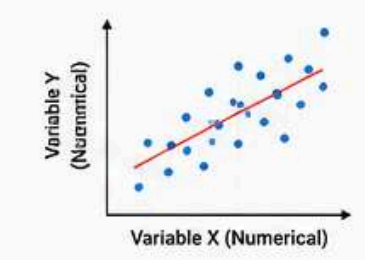
3 Need and Importance of Bivariate Analysis

- Helps understand interactions between two variables.
- Supports better business and research decisions.
- Identifies patterns and trends.
- Helps discover cause-and-effect relationships.
- Useful for predictive analysis.

4 Type 1 – Numerical vs Numerical

- Both variables are numerical.
- Used to study correlation and trends.
- Common tools:
 - Scatter Plot
 - Correlation Coefficient

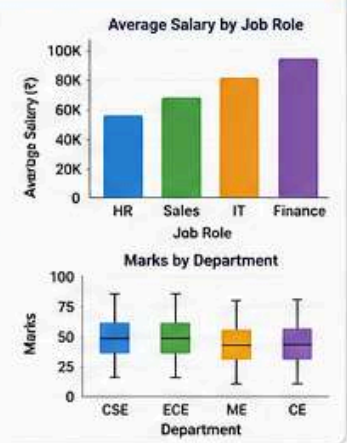
Example:
• Height vs Weight
• Income vs Expenditure



Objective:
Determine whether the variables are positively, negatively, or not correlated.

5 Type 2 – Numerical vs Categorical

- One variable is numerical and the other is categorical.
- Used to compare numerical values across different groups.
- Common tools:
 - Bar Charts
 - Box Plots



Example:
• Salary across different Job Roles
• Marks across different Departments

6 Type 3 – Categorical vs Categorical

- Both variables are categorical.
- Used to examine associations between categories.
- Common tools:
 - Contingency Tables
 - Chi-Square Test

Contingency Table Example (Gender vs Product Preference)

Gender	Product Preference		Total
	Product A	Product B	
Male	120	80	200
Female	90	110	200
Total	210	190	400

Chi-Square Test helps determine if there is a significant association between the two categorical variables.

Example:
• Gender vs Product Preference
• Education Level vs Employment Status

7 Applications of Bivariate Analysis

- Marketing Analysis
- Business Intelligence
- Educational Research
- Customer Behavior Analysis
- Sales and Revenue Studies

8 Multivariate Analysis

- Multivariate Analysis involves three or more variables simultaneously.
- Helps understand complex relationships among multiple variables.
- Widely used in advanced analytics, machine learning, and predictive modeling.



9 Example of Multivariate Analysis

Product Purchase Analysis

A business may study how:
• Age
• Income
• Education Level
together influence a customer's likelihood of purchasing a product. This provides deeper insights than analyzing only two variables.



10 Techniques of Multivariate Analysis

- Multiple Regression**
Examines the effect of multiple independent variables on one dependent variable.
- Principal Component Analysis (PCA)**
Reduces dimensionality while preserving important information.
- Cluster Analysis**
Groups similar observations into clusters.

11 Applications of Multivariate Analysis

- Customer Segmentation
- Risk Assessment and Modeling
- Disease Prediction
- Financial Analysis
- Market Research
- Machine Learning Applications



12 Difference Between Bivariate and Multivariate Analysis

Bivariate Analysis	Multivariate Analysis
Studies two variables	Studies three or more variables
Simpler analysis	More complex analysis
Focuses on one relationship	Focuses on multiple relationships
Example: Height vs Weight	Example: Age, Income, Education vs Purchase Decision

13 Conclusion

- Bivariate Analysis studies the relationship between two variables.
- Multivariate Analysis studies the interaction among three or more variables.
- Both techniques are important for discovering patterns, making predictions, and supporting data-driven decision-making.

Mean , Median and Mode

MEAN (Average)

DEFINITION

Mean is the sum of all observations divided by the total number of observations.

FORMULA

For Ungrouped Data:

$$\bar{X} = \frac{\sum_{i=1}^n X_i}{n}$$

Where,
 X_i = Each observation
 n = Total number of observations

For Grouped Data:

$$\bar{X} = \frac{\sum f_i X_i}{\sum f_i}$$

Where,
 f_i = Frequency of class
 X_i = Class mark (mid value)

EXAMPLE (Ungrouped Data)

Find the mean of: 2, 4, 6, 8, 10

$$\bar{X} = \frac{2 + 4 + 6 + 8 + 10}{5} = \frac{30}{5} = 6$$

Mean = 6

KEY POINTS

- Uses all the values in the dataset.
- Sensitive to extreme values (outliers).
- Best for symmetrical data.



MEDIAN (Middle Value)

DEFINITION

Median is the middle value in a dataset when the data is arranged in ascending or descending order.

FORMULA

For Ungrouped Data:

- If n is odd:
Median = Value of $\left(\frac{n+1}{2}\right)^{th}$ observation
- If n is even:
Median = $\frac{\text{Value of } \left(\frac{n}{2}\right)^{th} + \text{Value of } \left(\frac{n}{2} + 1\right)^{th}}{2}$

For Grouped Data:

$$\text{Median} = L + \left(\frac{\frac{N}{2} - c.f.}{f} \right) \times h$$

Where,

L = Lower boundary of median class
 N = Total frequency
 $c.f.$ = Cumulative frequency before median class
 f = Frequency of median class
 h = Class width

EXAMPLE (Ungrouped Data)

Find the median of: 3, 1, 7, 5, 9

Step 1: Arrange the data → 1, 3, 5, 7, 9

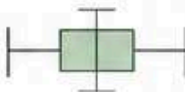
Step 2: $n = 5$ (odd)

$$\text{Median} = \text{Value of } \left(\frac{5+1}{2}\right)^{th} = \text{Value of } 3^{rd} \text{ observation} = 5$$

Median = 5

KEY POINTS

- Not affected by extreme values (outliers).
- Suitable for skewed data.
- Represents the middle position.



MODE (Most Frequent Value)

DEFINITION

Mode is the value that appears most frequently in a dataset.

FORMULA

For Ungrouped Data:

Mode = The value with the highest frequency

For Grouped Data:

$$\text{Mode} = L + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right) \times h$$

Where,

L = Lower boundary of modal class
 f_1 = Frequency of modal class
 f_0 = Frequency of class before modal class
 f_2 = Frequency of class after modal class
 h = Class width

EXAMPLE (Ungrouped Data)

Find the mode of: 4, 2, 2, 5, 7, 2, 8

Frequencies:

Value	4	2	5	7	8
Frequency	1	3	1	1	1

Mode = 2

KEY POINTS

- May not be unique (can have more than one mode).
- Not affected by extreme values.
- Useful for categorical data also.



QUICK COMPARISON

Mean – Uses all values, gives the average.
Best for: Symmetrical data



Median – Middle value, divides data into two halves.
Best for: Skewed data



Mode – Most frequent value.
Best for: Categorical data / Frequency analysis





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Thank you

